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0.1 刪除同字

```

1 #include<bits/stdc++.h>
2 using namespace std;
3
4 struct node{
5     int leng,idx;
6     bool operator<(const node& other) const {
7         if (leng!=other.leng) return leng<other.leng;
8         return idx>other.idx;
9     }
10 };
11
12 int main(){
13     int n;
14     cin>>n;
15     vector<int> tbl(n);
16     for(auto &i:tbl) cin>>i;
17     vector<int> num,len;
18     int siz=0;
19     num.push_back(tbl[0]);
20     len.push_back(0);
21     for(int i=0;i<n;i++){
22         if(tbl[i]==num[siz]) len[siz]++;
23         else{
24             siz++;
25             num.push_back(tbl[i]);
26             len.push_back(1);
27         }
28     }
29     siz=num.size();
30     vector<int> L,R,alive(siz,1);
31     for(int i=0;i<siz;i++){
32         L.push_back(i-1);
33         R.push_back(i+1);
34     }
35     priority_queue<node> pq;
36     for(int i=0;i<siz;i++) pq.push({len[i],i});
37     int ans=0;
38     while(!pq.empty()){
39         auto [currentL,idx]=pq.top();
40         pq.pop();
41
42         if(!alive[idx] || currentL!=len[idx]) continue;
43         ans++;
44         alive[idx]=0;
45         int l=L[idx],r=R[idx];
46         if(l>=0) R[l]=r;
47         if(r<siz) L[r]=l;
48
49         if(l>=0 && r<siz && alive[l] && alive[r] &&
50             num[l]==num[r]){
51             len[l]+=len[r];
52             alive[r]=0;
53             R[l]=R[r];
54             if(R[r]<siz) L[R[r]]=l;
55             pq.push({len[l],l});
56         }
57     }
58     cout<<ans;

```

57 }

0.2 滑窗

```

1 #include<bits/stdc++.h>
2 using namespace std;
3 multiset<int> l,r;
4 int tbl[200005];
5 void resort(){
6     while(l.size()<r.size()){
7         l.insert(*r.begin());
8         r.erase(r.begin());
9     }
10    while(l.size()>r.size()+1){
11        r.insert(*l.rbegin());
12        l.erase(prev(l.end()));
13    }
14 }
15 void insert_num(int num){
16     if(l.empty() || num<=*l.rbegin()) l.insert(num);
17     else r.insert(num);
18     resort();
19 }
20 void erase_num(int num){
21     if(l.find(num)!=l.end()) l.erase(l.find(num));
22     else r.erase(r.find(num));
23     resort();
24 }
25 int main(){
26     int n,x;
27     cin>>n>>x;
28     for(int i=0;i<n;i++) cin>>tbl[i];
29     for(int i=0;i<x;i++) insert_num(tbl[i]);
30     cout<<*l.rbegin()<<" ";
31     for(int i=x;i<n;i++){
32         insert_num(tbl[i]);
33         erase_num(tbl[i-x]);
34         cout<<*l.rbegin()<<" ";
35     }
36 }

```

0.3 滑窗 2

```

1 #include<bits/stdc++.h>
2 using namespace std;
3 #define int long long
4
5 signed main(){
6     ios::sync_with_stdio(0);cin.tie(0);
7     int n,k;
8     cin>>n>>k;
9     vector<int> a(n);
10    for(auto &c:a) cin>>c;
11    multiset<int> st;
12    for(int i=0;i<k;i++){
13        st.insert(a[i]);
14    }
15    auto mid=st.begin();
16    advance(mid,k/2-(k%2==0));
17    cout<<*mid<<" ";
18    for(int i=k;i<n;i++){
19        st.insert(a[i]);
20        if(a[i]<*mid) mid--;
21        if(a[i-k]<=*mid) mid++;
22        st.erase(st.lower_bound(a[i-k]));
23        cout<<*mid<<" ";
24    }
25 }

```

0.4 ai-x

```

1 #include<bits/stdc++.h>
2 using namespace std;
3 #define int long long
4
5 int n;
6 vector<double> arr;
7
8 double check(double val){
9     double cur_ma=0,ma=0;
10    double cur_mi=0,mi=0;
11
12    for(int i=0;i<n;i++){
13        double v=arr[i]-val;
14        cur_ma=max(0.0,cur_ma+v);
15        ma=max(ma,cur_ma);
16
17        cur_mi=min(0.0,cur_mi+v);
18        mi=min(mi,cur_mi);
19    }
20
21    return max(ma,-mi);
22 }
23
24 signed main(){
25     ios::sync_with_stdio(0);cin.tie(0);
26     cin>>n;
27     arr.resize(n);
28     for(auto &i:arr) cin>>i;
29     double l=-10000,r=10000;
30     int t=100;
31     while(t--){
32         double ml=l+(r-l)/3;
33         double mr=r-(r-l)/3;
34         if(check(ml)<check(mr)) r=mr;
35         else l=ml;
36     }
37     cout<<fixed<<setprecision(8)<<check(l)<<"\n";
38 }

```

0.5 圓形那題

```

1 #include<bits/stdc++.h>
2 using namespace std;
3 #define int long long
4
5 struct node{
6     double x,y;
7 };
8
9 int n;
10 vector<node> arr;
11
12 bool ok(double r){
13     double l=-1e18,rr=1e18;
14     for(int i=0;i<n;i++){
15         double delta=2*arr[i].y*arr[i].y*arr[i].y;
16         if(delta<0) return false;
17
18         double d=sqrt(delta);
19         l=max(l,arr[i].x-d);
20         rr=min(rr,arr[i].x+d);
21     }
22     return l<=rr;
23 }

```

```

24
25 signed main(){
26     ios::sync_with_stdio(0);cin.tie(0);
27
28     cin>>n;
29     arr.resize(n);
30
31     bool pos=0,neg=0;
32     for(auto &i:arr){
33         cin>>i.x>>i.y;
34         if(i.y>0) pos=1;
35         if(i.y<0) neg=1;
36     }
37
38     if(pos&&neg){
39         cout<<-1<<"\n";
40         return 0;
41     }
42
43     for(auto &i:arr) i.y=abs(i.y);
44
45     double l=0,r=1e15;
46     for(auto &i:arr) l=max(l,i.y/2);
47
48     for(int t=0;t<100;t++){
49         double mid=l+(r-l)/2;
50         if(ok(mid)) r=mid;
51         else l=mid;
52     }
53
54     cout<<fixed<<setprecision(10)<<r<<"\n";
55 }

```

0.6 計程車難

```

1 #include <bits/stdc++.h>
2 using namespace std;
3 #define int long long
4
5 struct carstatus{
6     int lasttime;
7     int dis;
8     int id;
9     bool operator>(const carstatus &other) const{
10         return lasttime > other.lasttime;
11     }
12 };
13
14 struct avlcar{
15     int lasttime;
16     int id;
17     bool operator<(const avlcar &other) const{
18         if(lasttime!=other.lasttime) return lasttime <
19             other.lasttime;
20         return id<other.id;
21     }
22 };
23
24 signed main(){
25     ios::sync_with_stdio(0);cin.tie(0);cout.tie(0);
26     int n,k,m;
27     cin>>n>>k>>m;
28     int tmp;
29     map<int,set<avlcar> > available;//dis lasttime id
30     for(int i=1;i<=k;i++){
31         cin>>tmp;

```

```

31     available[tmp].insert({0,i});
32 }
33
34 priority_queue<carstatus,vector<carstatus>,
35     greater<carstatus>> busy;
36 int t,f,e;
37 while(m--){
38     cin>>t>>f>>e;
39
40     while(!busy.empty() && busy.top().lasttime<=t){
41         available[busy.top().dis]
42             .insert({busy.top().lasttime,busy.top().id});
43         busy.pop();
44     }
45
46     if(available.empty()){
47         int near=busy.top().lasttime;
48         while(!busy.empty() &&
49             busy.top().lasttime==near){
50             available[busy.top().dis]
51                 .insert({busy.top().lasttime,busy.top().id});
52             busy.pop();
53         }
54
55         auto it=available.lower_bound(f);
56         vector<pair<int,avlcar>> cdd;
57         if(it!=available.end() && !it->second.empty()){
58             cdd.push_back({it->first,*it->second.begin()});
59         }
60         if(it!=available.begin() &&
61             !prev(it)->second.empty()){
62             cdd.push_back({prev(it)->first,*prev(it)->second.begin()});
63         }
64
65         pair<int,avlcar> best=cdd[0];
66         for(auto i:cdd){
67             int d1=llabs(best.first-f);
68             int d2=llabs(i.first-f);
69
70             if(d2<d1) best=i;
71             if(d2==d1){
72                 if(i.second.lasttime<best.second.lasttime)
73                     best=i;
74                 else
75                     if(i.second.lasttime==best.second.lasttime
76                         && i.second.id<best.second.id)
77                         best=i;
78             }
79         }
80
81         int wait=llabs(best.first-f);
82         if(best.second.lasttime>t)
83             wait+=llabs(best.second.lasttime-t);
84         int run=llabs(f-e);
85         cout<<best.second.id<<" "<<wait<<endl;
86
87         busy.push({t+wait+run,e,best.second.id});
88         available[best.first].erase(best.second);
89         if (available[best.first].empty())
90             available.erase(best.first);
91     }
92 }

```

```

1 struct node{
2     double x, y;
3 }arr[n];
4
5 double check(double val) {
6     double cmax = 0;
7     for(int i = 0; i < n; i++) {
8         cmax = max(cmax,(arr[i].x - 0) * (arr[i].x - 0) +
9             (arr[i].y - val) * (arr[i].y - val));
10    }
11    return cmax;
12 }
13
14 while(r - l > eps) {
15     double ml = l + (r - l) / 3;
16     double mr = r - (r - l) / 3;
17     if (check(ml) < check(mr)) {
18         r = mr;
19     }
20     else {
21         l = ml;
22     }
23 }

```